

Alex Morgan

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EXPERIENCE

Northwind Analytics

Austin, USA

Data Engineer

Jan 2024 – Present

- Developed components of a Medallion Architecture PySpark/Glue pipeline processing 3.5B+ daily records, storing outputs in Delta Lake format to power analytics, ML features, and reporting dashboards
- Onboarded 10+ data domains into enterprise Delta Lake tables within 6–8 day cycles by authoring SQL transformation configs and schema validation logic in Databricks
- Engineered serverless ingestion monitoring via Lambda, DynamoDB, and Step Functions (8-state workflow), automating delta file event triggering and improving pipeline observability
- Developed internal Lambda API tooling that automated pipeline configuration, testing, and deployment workflows, reducing engineering effort by ~70%
- Diagnosed and resolved 15+ high-severity data pipeline incidents via root cause analysis, eliminating recurring SLA breaches across 12 downstream data consumers
- Automated end-to-end test data provisioning by chaining 10+ Python and shell script steps across internal APIs and DynamoDB, cutting setup time ~75% and eliminating manual intervention
- Owned design and execution of 40+ end-to-end test scenarios across 3 integrated AWS Step Functions workflows, partnering with cross-functional product teams to translate stakeholder requirements into validated data quality checks

BlueOrbit Labs

Remote, USA

Data Engineering Intern

May 2023 – Aug 2023

- Designed an Airflow + dbt ETL pipeline loading 50M+ daily events from Kafka into Snowflake, cutting data-availability latency from 6 hours to under 30 minutes for 5 analytics teams

Acme AI

Remote, USA

Machine Learning Engineer Intern

May 2022 – Aug 2022

- Led team of 3 to build NLP preprocessing pipelines (TF-IDF vectorization, GloVe tokenization, BERT tokenization) for 40,000+ imbalanced support tickets across 5 intent classes
- Benchmarked Naive Bayes (TF-IDF), LSTM (GloVe 300-dim), and BERT (110M params) on 5-class intent classification, achieving 95.4% validation accuracy using TensorFlow and scikit-learn

SKILLS

Programming & Tools: Python, SQL, Linux/Unix, Git, Jenkins, Docker, Terraform, GitHub Actions

Databases: MySQL, Snowflake, MongoDB, PostgreSQL, Neo4j, BigQuery

Big Data & Visualization: PySpark, Spark, Delta Lake, dbt, Databricks, Airflow, Kafka, Hadoop, QuickSight

ML & AI: PyTorch, scikit-learn, HuggingFace Transformers, MLflow, BERT, LSTM, CNNs

AWS Services: Lambda, Glue, S3, DynamoDB, Redshift, Kinesis, IAM, EMR, Step Functions, Athena, EC2

PROJECTS

MarketPulse: Real-Time Streaming Analytics Platform

Jan 2024 – Mar 2024

- Built a Kafka + Spark Structured Streaming platform on AWS (MSK, EMR, S3, Redshift) processing 200K+ events/sec, surfacing sub-second operational dashboards in QuickSight with automated data quality checks on every micro-batch

CaptionAI: Image Captioning App with Audio

Sep 2023 – Nov 2023

- Designed an AI-driven image captioning system combining CNN & RNN models with text-to-speech capability, enhancing accessibility for visually impaired users and achieving a BLEU score of 0.21 using EfficientNet-B3

PUBLICATIONS

Adaptive Pooling Functions for Convolutional Neural Networks | *IEEE Example Conference 2022 – Best Paper Award*

- Implemented a novel adaptive pooling aggregation in CNN layers using PyTorch; outperformed classical pooling on state-of-the-art architectures across benchmark image datasets

ACCOMPLISHMENTS / VOLUNTEER WORK

Riverside State Datathon 2023: Secured 1st place, winning \$500 for predictive demand modeling of retail sales

Graduate Merit Fellowship: Awarded \$15,000 tuition fellowship in recognition of academic excellence

EDUCATION

Riverside State University <i>Master of Science in Data Science GPA: 3.9/4.0</i>	Austin, TX, US <i>Aug 2022 – May 2024</i>
Westbridge Institute of Technology <i>Bachelor of Technology in Computer Engineering</i>	Remote, US <i>Jul 2018 – May 2022</i>